

sophisticated software can process images for analysis.

Home security

Building or buying a house of your dream may not be enough, unless you safeguard it using appropriate home security systems. Choosing and buying home security systems require certain considerations. There are many alarm sensors, CCTV systems and other security solutions available in the market.

Motion sensors and CCTVs are the two most common home security systems. Motion sensors are usually installed outside the house. These let you know if someone is walking around your house or premise. Motion flood lights trigger lights when these sense motion. A motion guard dog sounds like the barking of a dog when someone approaches—speed, volume and type of the bark may vary depending on how far the detected motion is.

There are different types of motion sensors available in the market, including passive infrared (PIR), ultrasonic and tomographic.

PIR sensors. All warm-blooded



Fig. 3: Typical lidar on UAV (www.mdpi.com)

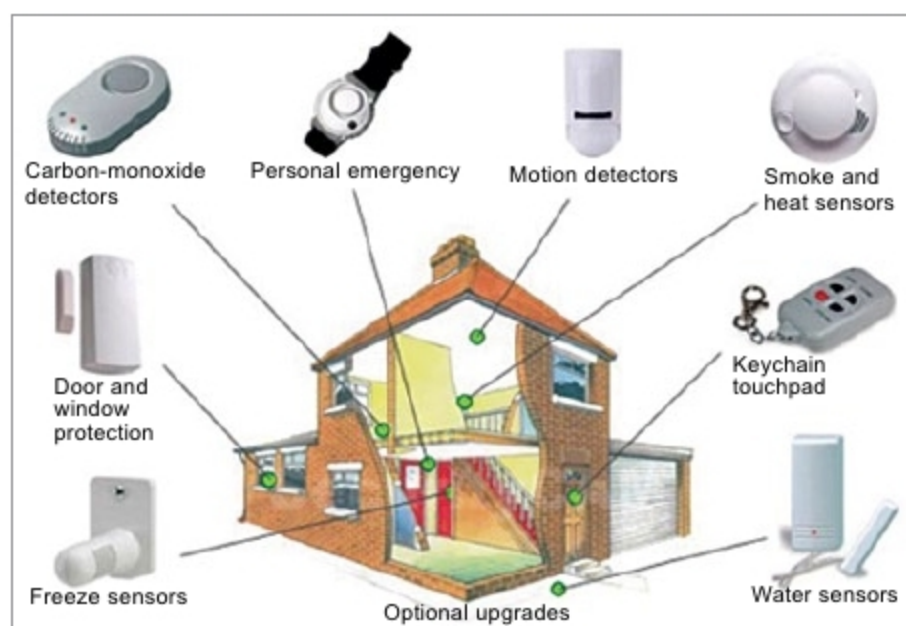


Fig. 4: Typical home security systems (<http://armetalarm.com>)

creatures including humans emit IR radiation. A PIR sensor triggers a burglar alarm when it detects a human or an animal. It is common in indoor alarms.

Ultrasonic sensors. These can be active or passive. Active ones emit pulses of ultrasonic waves and then measure reflected signals from a moving object. Animals can hear ultrasonic frequencies and so these signals may drive them away.

Tomographic sensors. These emit radio waves and sense when those waves are disturbed. These can detect waves through walls/objects, and are often positioned in a way that creates a radio wave around the area. These sensors are useful for warehouses and large storage units.

Smart home security systems. Smart security systems work in a seamless environment and can be manipulated using customised rules. These communicate with one or more wireless protocols such as Wi-Fi, Z-Wave, ZigBee or a proprietary mesh network.

A smart home security system connects to the home Wi-Fi network to monitor and control security devices using a smartphone. Using an app, you can monitor motion, windows, door locks, indoor and outdoor surveillance cameras, lights, sirens, smoke and more.

Personal security

Whether you are alone, at home, travelling, at your workplace or in a group, there can be many security issues.

Vulnerabilities and possibilities could come from smart TVs, computers, the Internet, smartphones and so on. The main goal of personal security alarms is to make a loud sound and attract attention in an emergency, and alert those nearby that the user is in danger. These gadgets are small, wireless, portable and easy to conceal.

Personal alarms are an excellent self-defence solution, especially for women and children. Some of these are small enough to fit in the palm of the hand or be attached to a keychain.

Panic alarms. These electronic devices can be attached to almost anything. The sensor unit activates a loud alarm and flash light. These devices are lightweight, compact and portable, and can be used to scare off potential intruders and prevent robberies.

Stun guns. Electronic stun devices often come with multiple features, including flash LED light and strobe function, and deliver a powerful electrical shock siren to deter threats. These may also have safety pin ropes that prevent others from using the devices against you.

Child locators. These wireless systems consist of transmitter and receiver units. Range of the transmitter is about 45.7m (150-feet) and can activate a sound alarm in the receiver unit. Most come with a wrist strap, which makes it easy to attach the receiver to shoelaces, belts or wrists.

Mobile SOS buttons. These are portable devices and can be worn around the neck by disabled people as a necklace or on the wrist as a bracelet. On pressing the button, these automatically call a number of landline or mobile numbers, one after another until the call is answered. The units then allow you to talk through the pendant or bracelet.

Implantable microchips. Use of microchips in livestock allows farmers to track animals. This technology can also be used for human beings. It can be particularly useful for parents who are concerned about the safety of their children. The microchip implanted in the child's body can transmit information related to the location of the child and alert parents of any danger.

Industrial security

Apart from CCTV systems, there are other security systems meant for industrial setups and related establishments. Due to the variety and complexity of commercial setups and industries, it is important to go beyond basic physical security, from simple to very complex systems. Emerging technologies, Industry 4.0 and Industrial